

KILPISJARVI SAANA, Finland

WMO: 027010

Lat: 69.043N

Lon: 20.851E

Elev: 1009

StdP: 89.78

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.5	(c) -19.8	(d) -25.0	(e) 0.4	(f) -22.0	(g) -22.2	(h) 0.6	(i) -18.9	(j) 25.5	(k) -10.2	(l) 23.5	(m) -11.4	(n) 11.2	(o) 140	(p) 1.117

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.9	(c) 18.2	(d) 12.5	(e) 16.0	(f) 11.8	(g) 13.9	(h) 10.4	(i) 13.5	(j) 17.0	(k) 12.3	(l) 15.0	(m) 11.1	(n) 13.3	(o) 6.3	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 12.1	(b) 9.9	(c) 13.9	(d) 11.0	(e) 9.2	(f) 12.9	(g) 9.8	(h) 8.5	(i) 12.1	(j) 41.0	(k) 17.0	(l) 37.6	(m) 15.2	(n) 34.5	(o) 13.1	(p) 17.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	19.1	16.9	-25.4	20.7	3.5	2.7	-27.9	22.6	-30.0	24.1	-31.9	25.6	-34.4	27.6
			-24.8	14.7	2.9	1.7	-26.9	15.9	-28.6	16.9	-30.3	17.9	-32.4	19.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.7	-11.3	-10.9	-9.4	-5.7	-0.7	4.2	9.0	7.3	3.3	-3.1	-6.8	-9.1		
	DBStd	8.01	4.54	4.62	3.46	3.62	4.66	4.27	4.22	3.58	3.59	3.55	3.68	3.57		
	HDD10.0	4706	661	586	602	470	332	182	71	98	201	407	505	592		
	HDD18.3	7683	919	819	861	720	589	426	289	341	451	665	755	850		
	CDD10.0	65	0	0	0	0	2	7	40	16	1	0	0	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.9	8.9	8.8	8.9	7.3	7.5	7.6	6.9	6.9	8.0	7.8	8.0	8.8		
(15) Precipitation	PrecAvg	497	52	39	34	24	27	43	68	46	36	41	36	51		
	PrecMax	617	151	90	159	89	51	83	132	98	64	80	78	103		
	PrecMin	355	19	7	8	4	7	4	17	20	9	6	3	18		
	PrecStd	61	30	18	35	16	12	19	28	18	17	20	20	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	-0.2	-0.9	6.0	16.0	17.9	22.5	18.6	13.5	7.4	5.2	0.1		
		MCWB	-0.8	-0.3	-1.4	2.6	9.0	11.9	14.6	13.3	6.9	2.4	2.3	0.1		
	2%	DB	-2.3	-2.4	-2.3	3.1	11.0	14.6	19.5	16.4	11.1	4.7	1.6	-1.8		
		MCWB	-2.5	-2.8	-2.6	1.3	7.3	10.4	13.1	12.0	7.5	3.2	1.1	-1.9		
	5%	DB	-3.8	-3.5	-3.5	0.2	8.2	12.2	17.4	14.2	9.3	3.3	-0.4	-3.2		
		MCWB	-4.0	-4.0	-3.7	-0.5	5.0	8.6	12.5	11.0	7.0	2.5	-0.6	-3.3		
	10%	DB	-5.1	-4.9	-4.6	-1.0	5.8	10.6	15.4	12.3	8.0	1.9	-2.1	-4.5		
		MCWB	-5.4	-5.5	-4.8	-1.4	3.4	8.0	11.4	10.1	6.4	1.1	-2.2	-4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	-0.2	-1.4	3.6	9.4	13.4	15.7	15.5	9.9	5.2	3.0	0.1		
		MCDB	-0.5	-0.1	-1.3	5.5	14.1	16.4	19.9	17.3	11.0	5.6	4.2	0.2		
	2%	WB	-2.3	-2.5	-2.6	1.6	7.4	11.2	14.0	12.7	8.4	3.5	0.8	-1.7		
		MCDB	-2.0	-2.4	-2.4	3.2	10.9	12.8	18.6	14.9	9.7	3.9	1.6	-1.7		
	5%	WB	-3.8	-3.7	-3.9	-0.1	5.2	9.9	12.9	11.4	7.4	2.4	-0.8	-3.2		
		MCDB	-3.6	-3.5	-3.7	0.4	8.0	12.2	16.6	13.5	8.7	3.1	-0.6	-3.1		
	10%	WB	-5.2	-5.2	-4.9	-1.3	3.6	8.0	11.9	10.3	6.5	1.2	-2.3	-4.7		
		MCDB	-4.9	-5.0	-4.7	-0.9	5.5	10.3	14.6	12.4	7.7	1.9	-2.1	-4.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.5	4.1	3.8	4.7	5.5	5.9	5.0	3.8	3.2	3.5	3.6		
		MCDBR	5.2	5.9	5.7	5.0	7.7	8.1	8.1	7.1	5.9	5.0	4.7	4.7		
	5% WB	MCWBR	5.2	6.0	6.0	4.1	4.5	4.8	3.9	4.2	4.1	4.0	4.2	4.8		
		MCWBR	5.1	6.3	5.6	4.8	7.6	7.3	7.1	6.2	5.1	4.4	4.7	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.162	0.212	0.256	0.278	0.293	0.299	0.304	0.292	0.266	0.224	0.163	N/A		
	taud		1.906	2.095	2.207	2.251	2.350	2.524	2.542	2.554	2.579	2.245	1.951	N/A		
	Ebn at Noon		65	578	779	871	898	905	887	861	798	564	53	0		
	Edh at Noon		5	41	73	96	98	86	82	72	53	34	4	0		
(44) All-Sky Solar Radiation	RadAvg		0.02	0.39	1.77	3.75	5.00	4.93	4.56	3.24	1.73	0.54	0.07	0.00		
	RadStd		0.00	0.04	0.10	0.20	0.40	0.40	0.40	0.25	0.16	0.06	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page